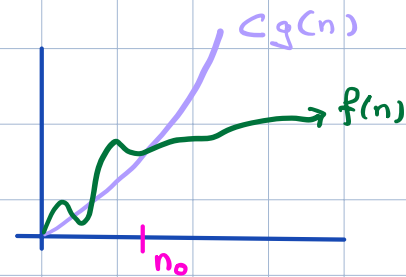


THE BIGS - CONTINUED!

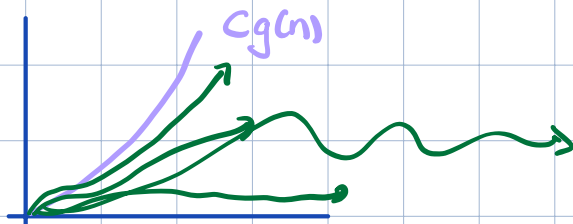
① Recall: $f(n) = O(g(n))$ means:

$\exists n_0, C > 0$ s.t. if $n \geq n_0$ then $f(n) \leq Cg(n)$



Intuition: $f(n)$ is eventually below (or equal to) a constant multiple of $g(n)$.

Issue: There's a lot of "freedom" in O in the following sense:



All are $O(g(n))$.

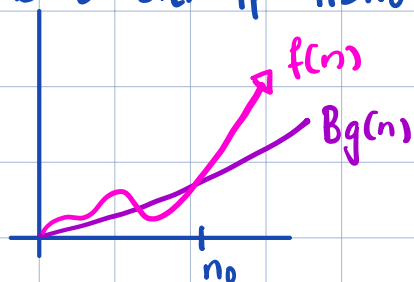
being O is not that restrictive.
It provides an upper bound

② Ω and Θ : (Omega and theta)

(A) Ω provides a lower bound in the following sense.

$\exists n_0, B > 0$ s.t. if $n \geq n_0$ then $f(n) \geq Bg(n)$ $\Leftrightarrow f(n) = \Omega(g(n))$

Pic



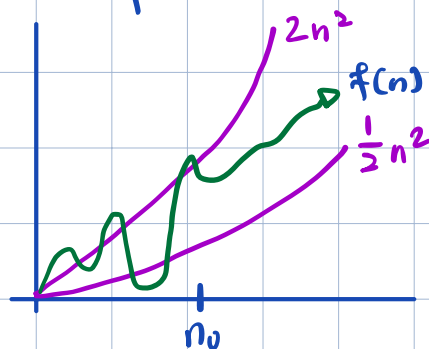
} eventually $f(n) \geq$ constant mult. of $g(n)$.

(B) Θ provides both! we write $f(n) = \Theta(g(n))$ if.

$\exists n_0, B > 0, C > 0$ s.t. if $n \geq n_0$ then $Bg(n) \leq f(n) \leq Cg(n)$

- Θ provides the most info!
- whenever possible we'll give Θ .

ex



Hoe $f(n) = \Theta(n^2)$

bk eventually it's b/w two mult. of n^2 .

FR: to prove Θ we typically prove O and Ω separately.

(by the defn). Won't be on Hw or Exam!

But we will prove Ω .

③ Proving $f(n) = \Omega(g(n))$.

Recall proving $f(n) = O(g(n))$:

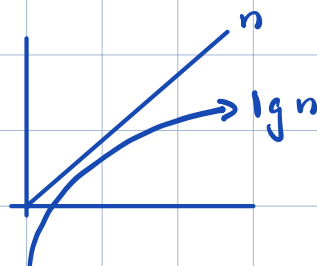
Prove $n^2 + n \lg n - n = O(n^2)$

well $n^2 + n \lg n - n \leq n^2 + n \lg n$ b/c $n \geq 0$ (assumed)

Then bound $n \lg n$ by a mult. of n^2 :

$n \lg n \leq n^2$ when $\lg n \leq n$

Which is always true:



thus if $n \geq 0$ then

$n^2 + n \lg n - n \leq n^2 + n^2 = 2n^2$ so defn sat. w/ $n_0 = 0$, $C = 2$

For Ω we have a similar procedure:

- we can drop +s.

ex Prove $n^2 \lg n + 8n^2 + n + 1 = \Omega(n^2 \lg n)$

well, $n^2 \lg n + 8n^2 + n + 1 \geq n^2 \lg n$ always.

So defn. sat w/ $B = 1$, $n_0 = 0$

- other exp. must be Conv. to mult. of the target fctn.

ex Prove $3n^2 - n = \Omega(n^2)$

note: want $3n^2 - n \geq \dots$ so n must be replaced
by a mult of n^2 which is $\geq n$.

well, $n^2 \geq n$ provided $n \geq 1$

so if $n \geq 1$ then

$$3n^2 - n \geq 3n^2 - n^2 = 2n^2$$

so defn. sat. w/ $B = 2$, $n_0 = 1$

BE CAREFUL!

ex Prove $n^2 - n = \Omega(n^2)$

if we try the same thing we get:

$$n^2 - n \geq n^2 - n^2 = 0 \quad \leftrightarrow \text{UH-OH! NOT A POS MULT OF } n^2.$$

can't "subtract it all away"!

sohn- don't!

For this problem obs: $\frac{1}{2}n^2 \geq n$ when $\frac{1}{2}n \geq 1$ so $n \geq 2$

In that case:

$$n^2 - n \geq n^2 - \frac{1}{2}n^2 = \frac{1}{2}n^2$$

so the defn is sat w/ $B = \frac{1}{2}$, $n_0 = 2$

• Could have used $\frac{1}{3}n^2 \geq n$ or $\frac{1}{4}n^2 \geq n$ or ...

goal is that some n^2 are left.

NOTE: In an ex. like $n^2 - n - n \lg n = \Omega(n^2)$

then something like $\frac{1}{3}n^2 \geq n$

$$\frac{1}{3}n^2 \geq n \lg n$$

b/c this ensures some n^2 are left over.

④ Intuition and Notes

(A) Typically we write $f(n) = \Theta(\text{nice fctn})$ (or O, Ω)
where "nice" means - like $1, n, \lg n, n^2, \dots$

However its also true that, for ex, $n^2 = O(n^2 + n \lg n + 1)$
we'd typically never do this.

(B) Our nice fctns are, inc. order:

$1, \lg n, n, n \lg n, n^2, n^2 \lg n, \dots, 2^n, 3^n, 4^n, \dots, n!, n^n, \dots$

Fun fact: all other logs are $\Theta(\lg n)$.

(C) Intuition says - given $f(n)$, find the largest ^{*} fctn
among its pieces and $f(n) = \Theta(\text{that})$.

So ex $n^2 + \underline{n^3 \lg n} - n + 1 = \Theta(n^3 \lg n)$

ex $\frac{1}{n} + \underline{4n^2} + \lg n - 1 = \Theta(n^2)$

note: if the largest is neg, take largest pos. bit:

ex $\underline{n} + \lg n - 10n^2 = \Theta(\underline{n})$